06. Model Selection

Explain why the notebook compares three baseline classifiers.

Project: CreditWise – Loan Approval Prediction

Source: supplied Loan-Approval-main.zip | Notebook: credit_wise.ipynb | Dataset: loan_approval_data.csv

Logistic Regression

A linear probabilistic classifier. It models log-odds as a linear combination of features: log(p/(1−p)) = β■ + β■x. It is a strong, interpretable baseline, but linear decision boundaries may underfit non-linear relationships unless feature engineering or interactions are added. Regularization is available through C and penalty settings.

K-Nearest Neighbours

KNN predicts based on the labels of nearby training observations. With n_neighbors=5, the notebook uses five nearest neighbours. It can model non-linear boundaries but is sensitive to scaling, irrelevant dimensions, distance metric, class imbalance and choice of K. Prediction can become costly as the training set grows.

Gaussian Naive Bayes

GaussianNB applies Bayes' theorem and assumes each feature follows a Gaussian distribution within each class and that features are conditionally independent given the class. It is fast and can work well as a baseline, but the independence assumption is often unrealistic for correlated financial variables.

Why these three

- Logistic Regression: interpretable baseline and probability output.
- KNN: non-parametric, distance-based comparison.
- GaussianNB: simple probabilistic model with strong assumptions.

Other candidates

- Decision Tree and Random Forest for non-linear interactions; tune depth and leaf sizes to control overfitting.
- Gradient Boosting / HistGradientBoosting or XGBoost/LightGBM if available, with careful calibration and validation.
- Explainable additive models or constrained models when policy interpretability is important.
- A scorecard/logistic model may be preferable in regulated lending contexts if it meets performance, governance and explainability requirements.

Selection caution

The notebook chooses GaussianNB based on the highest first-pass precision (0.804). After feature engineering, Logistic Regression has the highest F1 (0.810) and recall (0.836). The preferred model therefore depends on the chosen business objective; the repository does not establish one final model using a pre-defined cost function or robust cross-validation.